

Step-by-Step Guide: Creating a Public GitHub Repository with Multiple Collaborators

Introduction

If you want multiple people to work on the same GitHub project, the best approach is:

- Create a repository
- Add collaborators
- Each person clones the repository
- Everyone works on branches
- Changes are merged through Pull Requests

1 Method 1: Repository Owner Adds Collaborators

1.1 Step 1: Create a New Repository

1. Go to:

`https://github.com`

2. Click + → **New repository**

3. Fill in:

Repository Name: DAA-Project Description: Design and Analysis of Algorithms Project
--

4. Choose:

Public

5. Check:

Add README.md

6. Click:

Create repository

1.2 Step 2: Add Team Members

Open your repository.

Click:

Settings



Collaborators and teams



Add people

Search GitHub usernames:

john123
sarah456
alex789

Click:

Add collaborator

GitHub sends invitations.

1.3 Step 3: Collaborators Accept

Each member receives:

Invitation to collaborate

They click:

Accept Invitation

Now they can push directly.

1.4 Step 4: Everyone Clones Repository

Each member runs:

git clone https://github.com/OWNER/DAA-Project.git

Example:

git clone https://github.com/Yeabkalwo/DAA-Project.git

Enter project:

cd DAA-Project

1.5 Step 5: Create Personal Branch

Never work directly on main.

Example:

```
git checkout -b yeabkal-feature
```

Or:

```
git switch -c yeabkal-feature
```

1.6 Step 6: Make Changes

Create files:

```
algorithm.java  
README.md  
report.tex
```

Save your work.

1.7 Step 7: Commit Changes

Check status:

```
git status
```

Add files:

```
git add .
```

Commit:

```
git commit -m "Added Strassen algorithm implementation"
```

1.8 Step 8: Push Branch

```
git push origin yeabkal-feature
```

1.9 Step 9: Create Pull Request

Go to GitHub.

You will see:

```
Compare & Pull Request
```

Click it.

Add description:

```
Implemented Strassen matrix multiplication
```

Click:

```
Create Pull Request
```

1.10 Step 10: Review and Merge

Repository owner reviews.

Click:

Merge Pull Request



Confirm Merge

Now changes appear in:

main

2 Daily Workflow for Team Members

Before starting work:

```
git checkout main
git pull origin main
```

Create branch:

```
git checkout -b new-feature
```

Work.

Commit:

```
git add .
git commit -m "Added new feature"
git push origin new-feature
```

Create Pull Request.

3 If Someone Updated Main

Before pushing your work:

```
git checkout main
git pull origin main
```

Go back:

```
git checkout your-branch
```

Merge latest changes:

```
git merge main
```

Or:

```
git rebase main
```

Then push again.

4 Recommended Structure for Student Projects

DAA-Project

README.md
report
report.tex

source-code
Java
Python
C++

diagrams
figure1.png
figure2.png

references
papers.pdf

5 Example Team Setup

Suppose your team has 4 members:

Member	Branch
Yeabkal	yeabkal-feature
Samuel	samuel-feature
Ruth	ruth-feature
Roba	roba-feature

Each person works on their own branch and submits Pull Requests.

This prevents people from accidentally overwriting each other's work and is the workflow used by most professional software teams.